Got it—you want something even more structured: a **formal BS/MS/PhD-equivalent pathway**, but executed informally, with **clear academic mapping + weekly execution**.

I'll design this like a **top-tier university pipeline (BS → MS → PhD)** adapted for you.

🎓 INFORMAL AI DEGREE PATHWAY (BS → MS → PhD Equivalent)

Candidate Profile: Senior Engineer (18 yrs) **Track:** LLM Systems + AI Research **Total Duration:** 12 Months (Accelerated due to experience)

🧭 PROGRAM STRUCTURE (ACADEMIC EQUIVALENT)

Stage	Equivalent	Duration	Outcome
Stage 1	BS (Foundations)	3 months	Math + Core ML
Stage 2	MS (Specialization)	5 months	Transformers + Systems
Stage 3	PhD (Research)	4 months	Novel research + paper

📘 STAGE 1 — "BS LEVEL" (Weeks 1–12)

👉 Goal: **Strong theoretical + implementation base**

📚 Subjects Covered

- Linear Algebra
- Probability & Statistics
- Optimization
- Classical ML

📅 WEEK-BY-WEEK (BS LEVEL)

Weeks 1–2 → Linear Algebra

- Vectors, matrices, dot products
- Eigenvalues, SVD
- Implementation:
  - Matrix ops from scratch

Weeks 3–4 → Probability

- Distributions (Gaussian, Bernoulli)
- Expectation, variance
- Bayes theorem

Weeks 5–6 → Optimization

- Gradient descent
- Loss functions
- Convex vs non-convex

Weeks 7–8 → Neural Networks

- Forward pass
- Backpropagation
- Activation functions

👉 Build:

- 2-layer NN (NumPy only)

Weeks 9–10 → Core ML

- Logistic regression
- Decision trees
- Bias-variance

Weeks 11–12 → Mini Project

👉 Build:

- End-to-end ML pipeline

🎯 BS Deliverables

- NN from scratch
- ML pipeline
- Math clarity

📘 STAGE 2 — “MS LEVEL” (Weeks 13–32)

👉 Goal: **Deep specialization + systems thinking**

📚 Subjects Covered

- Transformers
- Deep Learning
- Distributed Systems
- LLM Architecture

📅 WEEK-BY-WEEK (MS LEVEL)

Weeks 13–16 → Transformers Core

- Attention mechanism
- Positional encoding
- Multi-head attention

👉 Build:

- Attention from scratch

Weeks 17–20 → GPT Architecture

Weeks 17–20 → GPT Architecture

- Decoder-only models
- Tokenization
- Training loop

👉 Build:

- GPT model using PyTorch

Weeks 21–24 → Training & Debugging

- Loss convergence
- Gradient issues
- Hyperparameter tuning

👉 Train model + debug instability

Weeks 25–28 → Systems & Scaling

- GPU usage
- Parallelism
- Memory optimization

Weeks 29–32 → Inference Systems

- KV cache
- Batching
- API serving

👉 Build:

- LLM inference system

🎯 MS Deliverables

- GPT implementation
- Training pipeline
- Inference system (production-style)

📘 STAGE 3 — “PhD LEVEL” (Weeks 33–52)

👉 Goal: **Research + contribution**

📚 Focus Areas

- Paper reading
- Reproduction
- Novel contribution

📅 WEEK-BY-WEEK (PhD LEVEL)

Weeks 33–36 → Paper Reading Phase

Sources:

- arXiv
- OpenAI

Tasks:

- 2 papers/week
- Re-derive math
- Summarize insights

Weeks 37–40 → Reproduction Phase

- Reimplement paper methods
- Validate results

👉 Output:

- 2 reproduction projects

Weeks 41–44 → Research Problem

👉 Choose one:

- Efficient transformer
- Faster inference
- Memory optimization

Weeks 45–48 → Experimentation

- Build prototype
- Run experiments
- Compare baselines

Weeks 49–52 → Paper Writing

- Write structured paper
- Publish on arXiv

🎯 PhD Deliverables

- Research project
- Paper (arXiv-level)
- Open-source repo

🧠 DAILY / WEEKLY STRUCTURE (ACADEMIC STYLE)

Weekday (2 hrs/day)

- 1 hr theory
- 1 hr implementation

Weekend (5–6 hrs/day)

- Deep work:
  - coding
  - experiments
  - paper reading

📊 EVALUATION RUBRIC

Level	You Can Do
BS	Implement ML from scratch
MS	Build GPT + systems
PhD	Publish research

🔥 WHAT MAKES THIS POWERFUL FOR YOU

Most people:

- Learn models ❌

You:

- Build + optimize + scale systems ✅

👉 Your edge: **AI + Architecture + Systems = rare profile**

⚠️ REALITY CHECK

This is equivalent to compressing:

- 4-year BS
- 2-year MS
- 4–5 year PhD

Into: 👉 **12 months (intense)**

So:

- No passive learning
- No skipping math
- No "just watching videos"

🚨 NEXT STEP (IMPORTANT)

🚀 NEXT STEP (IMPORTANT)

Don’t move forward blindly.

I strongly recommend next:

👉 I define **3 research problem statements tailored to your background** (these determine whether you reach PhD-level or not)

Just say: “**Give me research problems**”